Idiosyncratic Risk Dynamics in Indian Equity Markets During COVID-19

RESEARCH QUESTIONS

Primary Questions

- Does idiosyncratic risk increase along with market risk during crises (COVID-19)?
- Does idiosyncratic risk exhibit persistence (clustering) over time?

Additional Question

- How do co-movement and persistence of idiosyncratic risk vary across sectors?

MOTIVATION

Theoretical Expectation

- Idiosyncratic risk is firm-specific and diversifiable
- Should be uncorrelated with market risk

Breakdown During Crises

- Extreme events (e.g., COVID-19) may:
  - Increase firm-level uncertainty
  - Lead to co-movement with market risk
- Diversification benefits may weaken

Why This Matters

- If idiosyncratic risk co-moves with market risk and exhibits persistence, it challenges the foundational assumption that firm-specific risk is diversifiable

What We Study

- Co-movement with market risk
- Persistence over time
- Differences across sectors

Why COVID-19 & Time Period Design?

Why COVID-19?

- A global, exogenous shock to financial markets
- Period of extreme volatility and uncertainty
- Ideal setting to test breakdown of diversification

Why This Structure?

- Enables clean comparison across market regimes
- Captures:
- Normal conditions
- Stress period
- Recovery phase

Time Periods

- Pre-Crisis: Jan 2019 – Jun 2019
- Crisis: Feb 2020 – Jul 2020
- Post-Crisis: Jan 2023 – Jun 2023

Methodology Overview

1 Data Collection

- Daily adjusted closing prices (NIFTY 50 stocks)
- Risk-free rate: 91-day Treasury Bills

2 Return Construction

- Compute log returns
- Calculate excess returns (over risk-free rate)

3 CAPM Estimation

- Regress stock excess returns on market excess returns
- Obtain residuals (idiosyncratic risk component)

4 Risk Measurement

- Market and Idiosyncratic Risk: 30-day rolling variance of market returns and residuals

5 Regression Analysis

- Regression A: Idiosyncratic risk vs market volatility
- Regression B: Idiosyncratic risk vs lagged idiosyncratic risk

Data Description

Data Source

- Daily adjusted closing prices from Yahoo Finance
- Risk-free rate: 91-day T-Bill yields from RBI DBIE

Risk Proxies

- Market Risk:
- 30-day rolling variance of NIFTY 50 returns
- Total Risk:
- Average 30-day rolling variance of all selected stocks

Sample Selection

- NIFTY 50 stocks (large-cap, highly liquid)
- Excluded: M&M, TMPV, ETERNAL, Max Health
- Not traded across all selected time periods

Return Construction

- Log returns from adjusted prices
- Excess returns computed using risk-free rate
- Risk-free rate converted from auction yields to daily frequency

Returns, Excess Returns, Rolling Variance & Sector Classification

Returns

- Log returns computed from adjusted prices

$$R_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Excess Returns

- Returns adjusted for risk-free rate

$$R_i - R_f, \quad R_m - R_f$$

Rolling Variance

- 30-day rolling window used to estimate volatility
- Applied to:
  - Market returns → Market Risk
  - Residuals → Idiosyncratic Risk

$$\mathrm{Var}_t^{(30)} = \frac{1}{30-1}\sum_{i=0}^{29}\left(x_{t-i} - \bar{x}_t\right)^2 \qquad \bar{x}_t = \frac{1}{30}\sum_{i=0}^{29} x_{t-i}$$

Sector Classification

- Stocks grouped into sectors (e.g., Financials, IT, FMCG)
- Used to analyze:
  - Sensitivity to market risk
  - Persistence of idiosyncratic risk

CAPM REGRESSION

$$R_{i,t} - R_f = \alpha_i + \beta_i(R_{m,t} - R_f) + \epsilon_{i,t}$$

Implementation

- Regression run for each stock
- Residuals extracted as idiosyncratic component

```
y = excess_returns[stock]
X = market_excess
model = sm.OLS(y, X).fit()
```

Interpretation

- β: Sensitivity to market movements
- ϵ_i: Idiosyncratic (firm-specific) component

ESTIMATING IDIOSYNCRATIC RISK

Step 1: Residuals from CAPM

$$\epsilon_{i,t} = (R_{i,t} - R_f) - \hat{\beta}_i(R_{m,t} - R_f)$$

residuals[stock] = model.resid

Step 2: Rolling Variance (30-day window)

Idiosyncratic risk = variance of residuals

$$\sigma_{i,t}^2 = \frac{1}{30}\sum_{k=0}^{29}(\epsilon_{t-k} - \bar{\epsilon})^2$$

idio_risk = residuals.rolling(30).var()

Step 3: Aggregate Across Stocks

$$AvgIdio_t = \frac{1}{N}\sum_{i=1}^{N}\sigma_{i,t}^2$$

avg_idio = idio_risk.mean(axis=1)

REGRESSION DESIGN

Regression A: Co-movement with Market Risk

$$IdioRisk_t = \alpha + b \cdot MarketVol_t + \epsilon_t$$

- Tests whether idiosyncratic risk moves with market volatility
- b>0 → idiosyncratic risk increases during volatile periods

Regression B: Persistence (Clustering)

$$IdioRisk_t = \alpha + \rho \cdot IdioRisk_{t-1} + \epsilon_t$$

- Tests persistence of idiosyncratic risk over time
- ρ>0 → presence of clustering

Underlying CAPM Estimation

$$\hat{\beta} = \frac{Cov(R_i, R_m)}{Var(R_m)}$$

$$\hat{R}_i = \hat{\beta} R_m, \quad \epsilon = R_i - \hat{R}_i$$

Regression A Results

Regression 1: Idiosyncratic Risk vs Market Volatility

Period	Coefficient	P-value	R^2
Pre	-0. 0025	0.983	0
Crisis	0. 3692	0	0.899
Post	5.2074	0	0.314

Key Insights

- No significant relationship in the pre-crisis period
- Strong positive relationship during the crisis period
- Relationship remains positive in the post-crisis period

Implication

- Idiosyncratic risk increases with market volatility during crises
- Evidence of co-movement between firm-specific and market risk
- Suggests diversification weakens under market stress

Regression B Results

Regression B: Persistence of Idiosyncratic Risk

Period	Coefficient	P-value	R^2
Pre	0.9747	0	0.980
Crisis	0.9984	0	0.992
Post	0.9788	0	0.982

Key Insights

- Idiosyncratic risk is highly persistent across all periods
- Persistence increases further during crisis ($\rho \to 1$)
- Past risk almost fully explains current risk

Implication

- Strong evidence of volatility clustering
- Firm-specific shocks are long-lasting, not temporary
- Crisis periods amplify persistence effects

CROSS-PERIOD COMPARISON

Cross-Period Comparison: Co-movement & Persistence

Period	Co-movement (β)	Persistence (ρ)
Pre	Not significant	High (0.9747)
Crisis	Strong (0.3692)	Very High (0.9984)
Post	Strong (5.2074)	High (0.9788)

Key Patterns

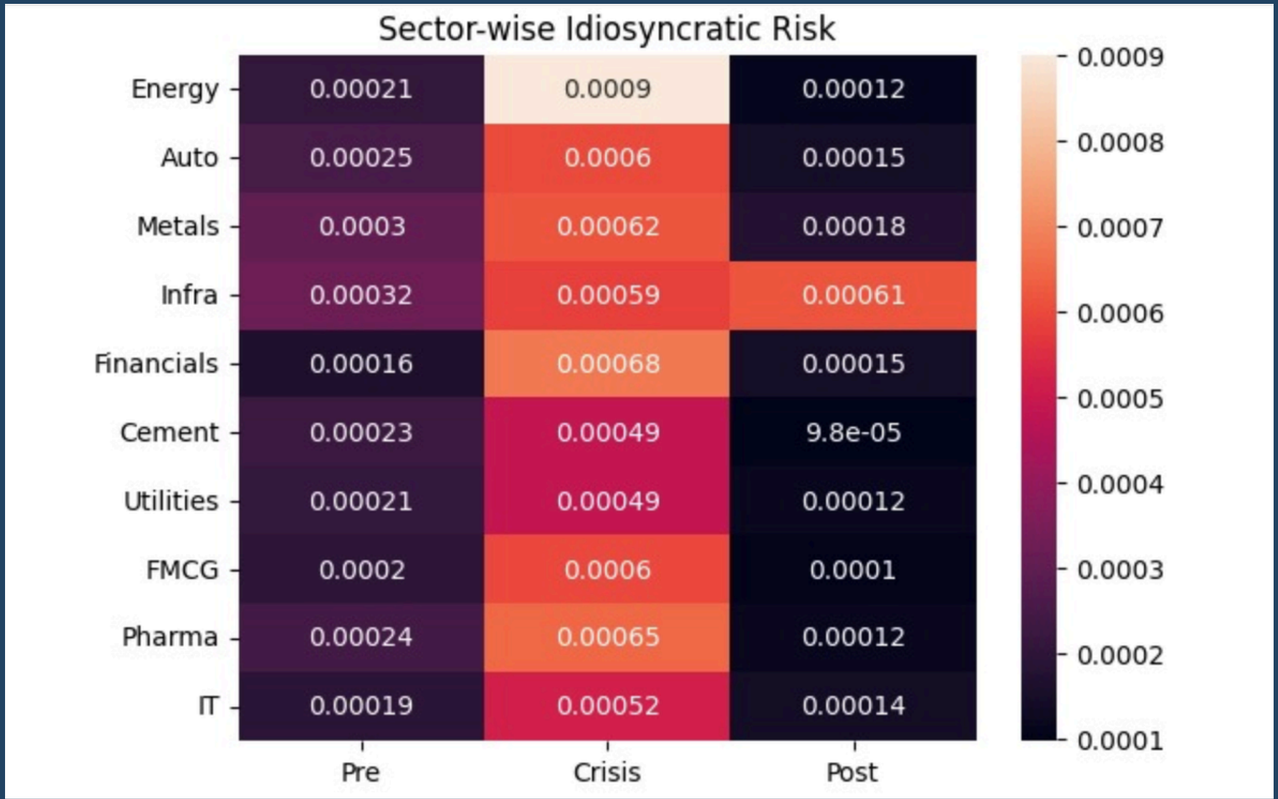
- Co-movement emerges only during crisis
- Persistence exists in all periods
- Crisis amplifies both effects significantly

Interpretation

- Pre-crisis: Idiosyncratic risk is largely independent
- Crisis: Firm-specific risk becomes systematically linked to market risk
- Post-crisis: Partial normalization, but effects persist

SECTOR-WISE ANALYSIS

Sector-wise Idiosyncratic Risk Across Periods



Key Observations

- Idiosyncratic risk increases across all sectors during crisis
- Energy and Financials show the highest spike
- Cement and Utilities remain relatively lower risk

Cross-Period Trend

- Pre-crisis: Low and stable across sectors
- Crisis: Sharp increase across all sectors
- Post-crisis: Risk declines but remains above pre-crisis levels

Implication

- Crisis impact is broad-based, not sector-specific
- However, magnitude varies → sector characteristics matter

SECTOR INSIGHTS

Sector Insights: Drivers of Idiosyncratic Risk

High Sensitivity Sectors

- Energy & Financials show the largest increase in risk
- More exposed to macro shocks and demand fluctuations
- Higher linkage to system-wide uncertainty

Moderate Sensitivity Sectors

- Auto, Metals, Infra show significant but moderate increases
- Driven by cyclical demand and supply chain disruptions

Lower Sensitivity Sectors

- Cement, Utilities, FMCG show relatively lower increases
- More stable demand and defensive characteristics

Key Takeaway

- Idiosyncratic risk response varies across sectors
- Sector characteristics influence vulnerability to crises

KEY FINDINGS

1. Co-movement Emerges During Crisis

- No relationship pre-crisis
- Strong positive relationship during crisis
- Idiosyncratic risk rises with market volatility

2. Strong Persistence Across All Periods

- Idiosyncratic risk is highly persistent ($\rho \approx 1$)
- Past risk strongly predicts current risk
- Evidence of volatility clustering

3. Crisis Amplifies Risk Dynamics

- Both co-movement and persistence increase significantly
- Market stress strengthens risk transmission

4. Sectoral Differences Matter

- Energy and Financials most affected
- Defensive sectors show relatively lower impact

Regression Analysis

- Idiosyncratic risk becomes partially systematic
- Benefits of diversification reduce under stress

INTERPRETATION

Breakdown of Diversification

- Idiosyncratic risk becomes correlated with market risk during crises
- Violates the assumption that firm-specific risk is fully diversifiable

Volatility Clustering in Firm-Specific Risk

- High persistence indicates shocks are long-lasting
- Firms do not revert quickly to normal risk levels

Market Stress Transmits to Firms

- System-wide shocks spill over into firm-level risk
- Blurs the distinction between systematic and idiosyncratic risk

Structural Shift in Risk Dynamics

- Post-crisis persistence and co-movement remain elevated
- Suggests lasting impact of extreme events on risk behavior

LIMITATIONS

Data Constraints

- Analysis limited to NIFTY 50 stocks.
- Excludes smaller firms and broader market dynamics.

Time Period Selection

- Focused on specific windows (Pre, Crisis, Post)
- May not capture long-term structural trends.

Limited Time Period:

- Relatively short time windows (6 months) may not fully capture the long-term risk dynamics.
- Findings based on a single crisis, limiting generalizability.

Risk Proxy Assumptions

- Market risk measured using 30-day rolling variance.
- May not fully capture all dimensions of volatility.
- Risk-free rate proxied using forward-filled T-bill yields → smoothens over day-to-day fluctuations in risk-free rate.

Limited Sector-wise analysis:

- Sector analysis is based on a limited number of large-cap firms, which may not fully represent sector dynamics.

FUTURE ROADMAP

Test Breakdown of Diversification

- Compute correlation across residuals.
- Compare Pre vs Crisis periods.
- Examine whether firm-specific risks become correlated.

Movement Between Regimes

- Observe transition dynamics, adjustment speed, lingering crises effects.

Refine Risk Measures

- Explore alternative volatility measures.
- Compare with realised volatility or GARCH-based estimates.

Sector-Level Deep Dive

- Analyse sector-specific drivers in more detail.
- Study heterogeneity within sectors and its implications.

Summary

- Idiosyncratic risk increases with market volatility during crises
- Risk is highly persistent, indicating volatility clustering
- Sector differences highlight varying exposure to shocks

Final Insight

- Idiosyncratic risk is not fully diversifiable during periods of market stress
- Crisis conditions blur the line between firm-specific and systematic risk

Thank you!